JavaScript String Methods

A simple guide to substring(), substr(), and slice()

1. First, understand character positions

JavaScript starts counting positions at **0**. So the first character is position 0, the second is position 1, and so on. Spaces also take a position.

Character	W	e	l	c	o	m	e	space	T	o
Index	0	1	2	3	4	5	6	7	8	9

Example text: `Welcome To Miracle`. The character at index 2 is **l**.

2. slice(start, end)

The **end** number is a position to stop at. The character at that end position is **not included**.

```
str.slice(2, 5)
```

Start at index 2 and stop just before index 5. That includes indexes 2, 3, and 4: **l, c, o**. Result: **"lco"**.

How to count slice

1. Put your finger on the start index.
2. Move right, including each character you pass.
3. Stop **before** the end index.

Memory tip: slice(start, end) means “from here up to there, but do not take the end position.”

Choose the right method

Page 2 - counting examples and important differences

3. substr(start, length)

The second number is the **number of characters to take**. Count the start character as number 1.

```
str.substr(2, 5)
```

Start at index 2, then take 5 characters total: **l (1), c (2), o (3), m (4), e (5)**. Result: **"lcome"**.

4. substring(start, end)

This behaves much like **slice()**: the start is included, and the end is excluded. For ordinary positive positions, the counting is the same.

```
str.substring(2, 5) // "lco"
```

A handy distinction: **substring()** swaps the two positions if start is greater than end. **slice()** does not; in that case it returns an empty string.

Side-by-side: Welcome To Miracle

Method	What the second number means	Example result
slice(2, 5)	Stop at index 5 (do not include it)	"lco"
substring(2, 5)	Stop at index 5 (do not include it)	"lco"
substr(2, 5)	Take 5 characters total	"lcome"

5. Quick rules to remember

slice: second number = stopping position; stop before it.
substring: second number = stopping position; stop before it.
substr: second number = how many characters to take.

Note for modern JavaScript

substr() is a legacy method and is not recommended for new code. Use **slice()** when you want a start and end position.

Try it yourself

For `"Welcome To Miracle"`, what does `slice(8, 10)` return? Start at index 8 and stop before index 10: the answer is "To".